Description

Illustration

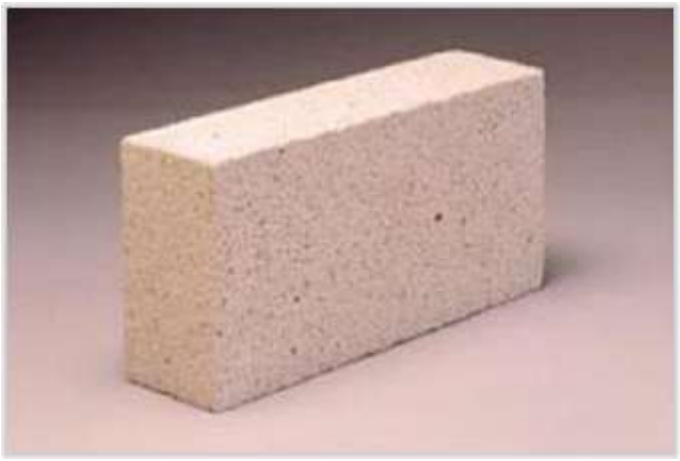




Figure caption

1. A single refractory brick, also known as fire brick. © ANSYS, Inc. 2. Refractory brick lining the inside of a drying oven. Image used under license from Shutterstock.com

The material

Refractory brick combines high thermal resistance with an ability to tolerate high temperatures. The porosity is high, trapping air pockets that provide the insulation. The alumino-silicate composition gives high-temperature strength. Refractory brick is used for kilns and furnaces and to line chimneys and fire-places.
The quoted properties are those for a single brick rather than that of brickwork, in which the properties and adhesion of mortar would have to be taken into consideration (most obviously becoming a dominant factor when loading in tension).

Compositional summary

30-40% alumina and 50% silica

General properties

Density	600	-	900	kg/m^3
Price	* 5.48	-	7.04	GBP/kg
Date first used	-7500			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	*	2	-	5	GPa
Shear modulus	*	0.8	-	2	GPa
Bulk modulus	*	1	-	4	GPa
Bending modulus	*	2	-	5	GPa
Poisson's ratio	*	0.18	-	0.22	
Yield strength (elastic limit)	*	0.75	-	0.95	MPa
Tensile strength	*	0.75	-	0.95	MPa
Compressive strength	*	4	-	5	MPa
Bending strength		0.8	-	1	MPa
Elongation	*	0.02	-	0.05	% strain
Hardness - Vickers	*	0.5	-	0.6	HV
Fatigue strength at 10^7 cycles	*	0.7	-	0.9	MPa
Fracture toughness	*	0.1	-	0.13	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	5e-4	-	0.001	

Thermal and combustion properties

Thermal conductor or insulator?		Good insulator			
Thermal resistivity		2.86	-	4	m.°C/W
Thermal expansion coefficient		4.5	-	6	µstrain/°C
Specific heat capacity		750	-	800	J/kg.°C
Melting point	*	1.73e3	-	1.84e3	°C
Maximum service temperature		1.28e3	-	1.38e3	°C
Minimum service temperature		-273			°C
Flammability		Non-flammable			
Emissivity		0.75	-	0.93	

Hygro-thermal properties

Water absorption		0.2	-	0.3	%
Frost resistance		Average			

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	1e17	-	1e19	µohm.cm
Dielectric constant (relative permittivity)	*	4	-	10	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.01	
Dielectric strength (dielectric breakdown)	*	5	-	7	MV/m

Optical properties

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	Yes			

Acoustic properties

Sound absorption	Good			
Sound isolation	Average			

Processability (scale 1 = impractical to 5 = excellent)

Moldability	2	-	4	
Machinability	1	-	2	

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Excellent
Weak alkalis	Acceptable
Strong alkalis	Limited use
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Limited use
Industrial atmosphere	Limited use
Rural atmosphere	Acceptable
Marine atmosphere	Acceptable

Geo-economic data for principal component

Annual world production, principal component	5e7	-	5.1e7	tonne/yr

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.188	-	0.208	kg/kg
Embodied energy (virgin grade production)	* 3.27	-	3.61	MJ/kg
Water Usage	* 21.8	-	24.1	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 16.5	-	17.9	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	*	1.08	-	1.17	kg/kg

Recycling and end of life

Recycle	✓				
Functional recycle	✓				
Climate change (CO2-eq), recycling	*	0.0188	-	0.0208	kg/kg
Embodied energy, recycling	*	0.458	-	0.505	MJ/kg
Recycle fraction in current supply		0			%
Combust for energy recovery	✗				
Landfill	✓				
Biodegrade	✗				
A renewable resource?	✗				

Supporting information

Design guidelines

Bricks and the mortar used to bond them are strong in compression but weak in tension. This largely determines the way they are used: brick structures (like those of stone) are designed in such a way that the loads are compressive everywhere, ruling out cantilevered or very slender forms. The face of brick can be molded and glazed, allowing great freedom of decoration, and the pattern and natural color variation gives a visually attractive surface. Low -fired, unglazed brick is vulnerable to water-penetration and degradation, and should be screened from direct weathering. High-fired or glazed brick is extremely durable.

Technical notes

Brick is fired clay: hydrous aluminum silicate, with impurities of potash, soda, lime, and oxides of iron. On heating to 900-1200C, the aluminum silicate reacts with soda to form a viscous glass - it is this that bonds the brick together. Most brick is porous; if it is to resist water it is glazed. To do this the surface of the fired brick is painted with a mixture of glass-forming fluxes and the brick is re-fired, melting the glaze, and forming a glassy surface that can be colored. Large pores and low porosity are the most important factor in establishing if a particular brick type will be durable. The quoted properties are those for a single brick rather than that of brickwork, in which the properties and adhesion of mortar would have to be taken into consideration (most obviously becoming a dominant factor when loading in tension).

Typical uses

Thermal insulation for kilns, furnaces; linings for chimneys and fire-places.

Tradenames

CSI/CSC Masterformat Section Numbers:
04 05 16 Masonry Grouting
04 05 23 Masonry Accessories
04 20 00 UNIT MASONRY
04 21 00 Clay Unit Masonry
04 23 00 Glass Unit Masonry
04 50 00 REFRACTORY MASONRY
04 51 00 Flue Liner Masonry
04 57 00 Masonry Fireplaces

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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